

CHRONON-1

Statistical Analysis Plan (SAP) / Preregistration

Metrology preregistration (public datasets): optical-clock redshift/LPI null test

Execution contract: endpoints, data, preprocessing, primary models, gates, decision rules, reproducibility

v1.0 — 05/01/2026

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Host: (not assigned at publication time)

Protocol reference (CHRONON-1). CHRONON-1: Experimental Pre-Registration (program protocol) (v1.0 — 10/11/2025).

DOI: 10.5281/zenodo.17571491

Execution reference (replication package). Replication package archived on Zenodo. Concept DOI (all versions): 10.5281/zenodo.18072279. Version DOI (v1.0): 10.5281/zenodo.18072280.

Note. This preregistration is published before PI/Host assignment and does not imply endorsement by any institution or individual.

One-line pitch. Falsifiability-first program: preregistered null tests with explicit anti-systematics gates, producing a single primary output (robust bound or methodological no-go).

Data policy. Public/archived datasets only (DOI/URL + immutable version + checksum). No internal access; no sensitive or personal data.

Key principle. One primary endpoint + one primary model per test; everything else is sensitivity or exploratory, never confirmatory.

Freeze-and-publish. No primary run at publication time. Execution requires frozen code/config and a completed `MANIFEST.json` + `FREEZE.txt`. Any change implies a new version (v1.1+).

PI/HOST ONE-PAGE (QUICK READ)

Goal (Test A). A preregistered null-test on optical-clock comparisons: estimate a *single* GR-subtracted residual slope $\hat{\epsilon}_\Phi$ versus gravitational exposure $X \equiv \Delta U/c^2$ (or $g \Delta h/c^2$). Output is either a robust bound (EEP/LPI-style) or a methodological no-go — both publishable.

What I ask from the PI (light load).

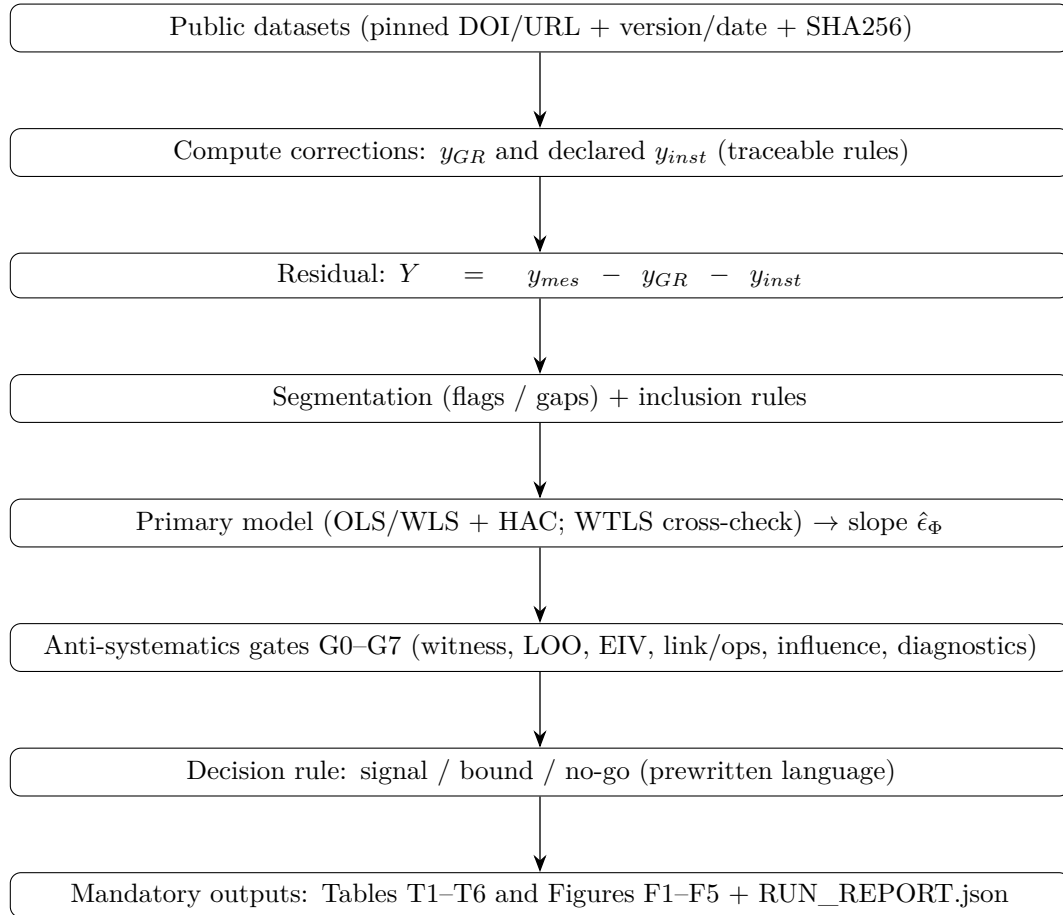
- Sanity-check endpoints, gate thresholds, and the dominant systematic risk.
- Sign-off the freeze record (`FREEZE.txt`) once the sources + code are pinned.
- Quarterly 30-min check-in (remote OK).

What I provide (heavy lifting on my side).

- Deterministic pipeline + one-command replication package (`run_all`).
- Prewritten decision language (signal / bound / no-go) locked to gates.
- Manuscript + public artefacts + reviewer responses.

What I do *not* ask. No ontological endorsement, no internal data access, no lab hardware commitment.

Analysis chain (one glance).



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I — PURPOSE AND SCOPE

This document is. An execution contract: operational definitions, data provenance, deterministic pipeline, primary model, gates, decision rules, mandatory outputs, and reproducibility.

This document is not.

- A results paper (no primary estimate is computed at freeze time).
- A hardware description (hardware stays in the main protocol; this SAP covers the analysis chain only).
- A license to post-hoc reinterpret the outcome: any non-pre-registered analysis is explicitly labelled exploratory.

Confirmatory vs exploratory.

- **Confirmatory (pre-registered):** Sections I–XIII (endpoints, data, pipeline, models, gates, decision).
- **Exploratory:** anything outside the declared sensitivity set in Section IX.

Key rule (anti-p-hacking). Exactly one primary endpoint and one primary model per test. Secondary results never upgrade a failed primary gate into a claim.

II — SCIENTIFIC QUESTIONS AND TESTS

This SAP supports independent tests (work packages). Each test has: (i) one primary endpoint, (ii) one primary model, (iii) gates, (iv) a decision rule.

II.1 Test A (optical clocks) — GR-subtracted residual slope

Question. After GR correction, is there a signed residual slope ϵ_Φ correlated with gravitational exposure?

Null / alternative. $H_0 : \epsilon_\Phi = 0$. $H_1 : \epsilon_\Phi \neq 0$ (two-sided).

II.2 Test B (qubits, optional) — decoherence / tempo exposure

Question. Does a qubit coherence observable exhibit a monotone dependence on an operational tempo exposure X_Φ (defined in Section V)?

Null / alternative. $H_0 : \alpha_\Phi = 0$. $H_1 : \alpha_\Phi \neq 0$ (two-sided).

Non-negotiable rule

If a test is not executed (e.g., Test B), it is explicitly marked **NOT EXECUTED**. No claim is allowed from partial outputs.

III — ENDPOINTS AND CLAIM CONTROL

III.1 Primary endpoint (Test A)

Exposure. $X \equiv g \Delta h / c^2$, equivalently $X = \Delta U / c^2$ when a geopotential model is used. Sign convention is fixed: $\Delta h \equiv h_B - h_A$ and $X \equiv X_B - X_A$.

GR-subtracted residual. For each observation i (timestamp + pair/path + configuration),

$$Y_i \equiv y_i^{\text{res}} = y_i^{\text{mes}} - y_i^{\text{GR}} - y_i^{\text{inst}},$$

with $y^{\text{mes}} = \Delta f / f$ (dimensionless), y^{GR} the reference GR correction, and y^{inst} the sum of declared instrumental corrections (link, jumps, drift, calibrations).

Endpoint. $\hat{\epsilon}_\Phi$: estimated signed slope of Y versus X under the primary model (Section VIII). The primary report includes $\hat{\epsilon}_\Phi$, its HAC standard error, a 95% CI, and the full gate table.

III.2 Primary endpoint (Test B, optional)

Observable. Default is $Q \equiv T_2^{-1}$ (as in the main protocol summary). If the dataset provides T_1 and T_ϕ , the decomposition is treated as diagnostic; the primary endpoint remains a single slope.

Exposure. X_Φ must be computable from public information and must be fixed before freeze (Section V).

Endpoint. $\hat{\alpha}_\Phi$: slope of Q versus X_Φ under the primary model.

III.3 Multiplicity and allowed claims

- One primary claim per test, based **only** on the primary endpoint + primary model + gates.
- Declared sensitivities are reported as robustness diagnostics. They cannot rescue a failed gate.
- Any additional analysis is exploratory and cannot support a confirmatory claim.

IV — DATA SOURCES, TRACEABILITY, AND INTEGRITY

Rule. Every source is pinned by DOI/URL, immutable version/date, and SHA256 checksum verified at ingestion.

Eligibility rule

If a listed source becomes unavailable after freeze, it is excluded (no post-hoc replacement), unless an *identical* mirror exists with the same checksum.

IV.1 Declared sources (example structure)

The final freeze includes a machine-readable `MANIFEST.json` (Annex E) listing all sources and checksums.

Test	Source (DOI/URL)	Version/date	SHA256
A	(public dataset #1)	YYYY-MM-DD	(hash)
A	(public dataset #2)	YYYY-MM-DD	(hash)
B	(public dataset #1)	YYYY-MM-DD	(hash)

V — VARIABLE DICTIONARY, UNITS, AND SIGNS

V.1 Time and conventions

- Time is UTC by default. If the source uses TAI/GPS/TT, conversion is applied at ingestion with a versioned leap-second table.
- All dimensional quantities are converted to SI. Frequency fractions are dimensionless.
- Sign: $X \equiv X_B - X_A$ and Y follows the dataset convention for $\Delta f/f$; any conversion is explicitly logged.

V.2 Test A variables (optical clocks)

- $y^{\text{mes}}(t) \equiv \Delta f/f(t)$.
- $X(t) \equiv g \Delta h(t)/c^2$ and/or $X(t) = \Delta U(t)/c^2$ with $\Delta U(t) = U_B(t) - U_A(t)$.
- $y^{\text{GR}}(t)$: reference GR correction. Minimum: $X(t)$. Additional terms (kinematic, tides) are included **only if** they are traceable from public metadata; otherwise they are omitted and documented.
- $y^{\text{inst}}(t)$: declared instrumental corrections, each either provided as metadata or reconstructed from a deterministic rule (Section VII).
- $Y(t) = y^{\text{mes}} - y^{\text{GR}} - y^{\text{inst}}$.
- $\sigma_X(t)$: declared (or derived) uncertainty on $X(t)$ from public levelling/geopotential metadata. If time-varying geopotential terms (tides/loading) cannot be computed from public inputs, they are set to zero and absorbed into a documented conservative σ_X (used in Section VIII and reported in T1/T3).
- `link_id` / `path_id` (if provided): identifier(s) used for leave-one-link-out diagnostics and gates.
- Quality / operations flags (if provided): `flag_jump`, `flag_cycle_slip`, `flag_unlock`, `flag_maintenance`. Any flagged interval defines a segment boundary and is excluded in the primary run (Section VI).

V.3 Allowed covariates (closed list)

Covariates are allowed **only if** (i) public with the data, (ii) defined here, (iii) used as declared in gates/sensitivities.

- $Z_{\text{temp}}(t)$: temperature (K).
- $Z_{\text{hum}}(t)$: humidity (%).

- $Z_{\text{link}}(t)$: link quality metric (SNR / loss / etc.).
- $Z_{\text{tide}}(t)$: tide correction term (if provided).
- Z_{config} : discrete configuration indicator (encoding fixed).

V.4 Test B variables (qubits, optional)

- $Q(t) \equiv T_2^{-1}(t)$ (default).
- $X_{\Phi}(t)$: operational tempo exposure (fixed before freeze). Allowed choices:
 - $X_{\Phi} = \Delta \ln \Phi$ between two conditions/locations, or
 - a public proxy (altitude / geopotential / etc.) mapped to $\Delta \ln \Phi$ through a declared rule.

VI — ANALYSIS DATASET CONSTRUCTION

VI.1 Unit of analysis

Long-table format: one row per observation $i = (t_i, \text{pair}, \text{siteA}, \text{siteB}, \text{config})$. Required fields: t_i (UTC), pair_id , y_i^{mes} , and enough metadata to compute X_i . Optional: covariates, link metrics, quality flags.

VI.2 Segmentation (a priori rules)

Goal: isolate quasi-stationary windows (stable configuration, no unmodelled jump).

Segmentation parameters (locked)

Gap threshold: $\tau_{\text{gap}} = 24$ h. Minimum segment length: $T_{\text{min}} = 48$ h.

Rules (in order):

1. Explicit boundaries: if the dataset provides a jump/maintenance flag, it defines a segment boundary.
2. Temporal gaps: a gap $> \tau_{\text{gap}}$ splits segments.
3. Minimum duration: segments with duration $< T_{\text{min}}$ are excluded and logged.
4. Unflagged step audit (diagnostic): a deterministic change-point detector is run on y^{mes} and on Y . Any detected step above the locked threshold that is not aligned with declared flags is logged and evaluated by Gate G6.

Each segment receives a stable `segment_id`.

VI.3 Inclusion/exclusion (mechanical)

Include if all hold: (i) source listed and checksum verified, (ii) required variables present, (iii) valid segments. **Exclude** if any holds: missing minimal metadata (cannot compute X or Y), unit inconsistency not resolvable, short/invalid segment, or explicit quality flags (maintenance/invalid/saturation). Every exclusion generates a log line.

VI.4 Declared subsets

Witness subset ($\Delta h \simeq 0$). Define X_0 as the 5th percentile of $|X|$ over the full analysis dataset. Witness set: $\{i : |X_i| < X_0\}$. (Data-dependent threshold is allowed because the rule is fixed here.)

Independence splits. (i) Temporal split into $K = 3$ disjoint blocks. (ii) Leave-one-segment-out (LOO) for gate ablation. (iii) Leave-one-link-out when `link_id/path_id` exists.

VII — PREPROCESSING (DETERMINISTIC PIPELINE)

Principle. Minimal, traceable, deterministic. No “visual tuning”.

VII.1 Pipeline steps (fixed order)

1. Ingestion: download $\rightarrow$ checksum verification $\rightarrow$ parse to canonical table.
2. Normalisation: units, UTC time, column names.
3. Exposure computation: X (and X_{Φ} if Test B).
4. Corrections: compute y^{GR} , then y^{inst} .
5. Residual: compute $Y = y^{\text{mes}} - y^{\text{GR}} - y^{\text{inst}}$.
6. Segmentation: apply Section VI rules.
7. Link/ops QC: apply exclusion rules for flagged events, compute link-quality summaries (if available), and run the unflagged step audit (Section VI; Gate G6).

8. Minimal cleaning: outlier flagging (declared rule below) **without deletion** in the primary run.
9. Diagnostics: produce the mandatory figures/tables (Section XIV).

VII.2 Integrity checks (must pass)

- Schema check: required columns present, types valid, units declared.
- Range checks: X order-of-magnitude sanity (e.g. $|X| \lesssim 10^{-15}$ for terrestrial Δh).
- Time monotonicity and duplicates.
- Checksum audit: all source files match `MANIFEST.json`.

VII.3 Outlier flagging (declared, deterministic)

Outliers are flagged (not removed) by robust MAD rule on windowed residuals:

$$\text{flag_outlier}_i \Leftarrow |r_i - \text{median}(r)| > \kappa \cdot \text{MAD}(r), \quad \kappa = 5.$$

Any deletion is restricted to declared sensitivity runs only (Section IX).

VII.4 Blinding transform (development only)

During development, the real-data residual used for any fit is replaced by an affine-blinded version:

$$Y^{(\text{blind})} = aY + b,$$

with $a \neq 0$ and b fixed by a seeded draw and stored in `config.yaml`. This prevents tuning to the true slope.

VIII — PRIMARY MODELS

VIII.1 Test A primary model

Free-intercept linear model:

$$Y_i = \alpha + \epsilon_\Phi X_i + \beta^\top Z_i + \varepsilon_i.$$

Estimator:

- Default: WLS with weights $w_i = 1/\sigma_{Y,i}^2$ when $\sigma_{Y,i}$ is provided; else OLS.
- Standard errors: HAC (Newey–West) with Andrews bandwidth (deterministic rule).
- Exposure-uncertainty / errors-in-variables (locked): define $r_X \equiv \text{median}(\sigma_X)/\text{span}(X)$ (with σ_X from public metadata or a conservative construction, Section V). If $r_X \geq 0.02$ or $\sigma_X/X_{\text{rms}} \geq 0.1$, run Deming/WTLS with $\lambda = \sigma_Y^2/\sigma_X^2$ fixed from propagated uncertainties. This run is a mandatory robustness check and is enforced by Gate G3.

VIII.2 Test A primary test statistic and threshold

Primary statistic:

$$t \equiv \frac{\hat{\epsilon}_\Phi}{\text{SE}_{\text{HAC}}(\hat{\epsilon}_\Phi)}.$$

Confirmatory threshold (locked to CHRONON-1): two-sided 5σ , i.e. $|t| \geq 5$. The 95% CI is reported regardless.

VIII.3 Test B primary model (optional)

$$Q_i = \alpha_\Phi + \alpha_\Phi X_{\Phi,i} + \beta^\top Z_i + \varepsilon_i,$$

with the same WLS/HAC rules. If $Q = T_2^{-1}$ is strongly skewed, the alternative transform $Q = \ln T_2$ is allowed only as a declared sensitivity (Section IX).

IX — DECLARED SENSITIVITY ANALYSES

Purpose. Robustness diagnostics only. No confirmatory claim is allowed from sensitivities.

IX.1 Model sensitivities (Test A)

1. No covariates: drop Z .
2. No fixed effects (if used): global fit without site/operator indicators.
3. GLS AR(1) per segment (always reported as a robustness diagnostic).

4. Robust regression (Huber).
5. HAC bandwidth sensitivity: half and double the Andrews bandwidth.
6. Block bootstrap (moving blocks) CI cross-check with deterministic block length (reported; not confirmatory).

IX.2 Sample sensitivities (Test A)

1. Leave-one-segment-out (LOO): recompute $\hat{\epsilon}_{\Phi}^{(-s)}$ for each segment.
2. Leave-one-site-out (if applicable).
3. Leave-one-link-out (if `link_id/path_id` is available).
4. Outlier ablation: exclude flagged outliers (and only those) and rerun.
5. Link/ops ablation (if flags/metrics exist): exclude flagged link events (cycle slips/unlocks/maintenance) and rerun.
6. Temporal split: run separately on the $K = 3$ disjoint blocks.

IX.3 Outputs

A table reports $\hat{\epsilon}_{\Phi}$, SE, CI for each sensitivity, plus diagnostics: sign stability, amplitude shifts, influential segments.

X — DECISION RULES AND ALLOWED LANGUAGE

Primary claim logic (Test A)

A “signal” claim is allowed **only** if: (i) $|t| \geq 5$ on the primary run, **and** (ii) **all gates** in Section XI are PASS, **and** (iii) no major deviation is declared (Section XV).

X.1 Pre-written conclusions (Test A)

Case A (all gates PASS and $|t| \geq 5$). “We observe a non-zero signed GR-subtracted slope $\hat{\epsilon}_{\Phi}$ correlated with X , with HAC uncertainty and a 95% CI reported. All pre-registered anti-systematics gates are passed; the result is operationally compatible with a robust signal under CHRONON-1.”

Case B (all gates PASS and $|t| < 5$). “No significant slope is detected at the pre-registered 5σ threshold. We report a robust bound on $|\epsilon_{\Phi}|$ from the 95% CI.”

Case C (any gate FAIL). “The test is not valid for confirmatory inference (failed gate(s): list). We report diagnostics and sensitivity results only. No EEP/LPI/tempo claim is made.”

X.2 Secondary/exploratory language

Any statement based on sensitivities is labelled “sensitivity”; any additional analysis is labelled “exploratory” and cannot be used for confirmatory inference.

XI — GATES (ANTI-SYSTEMATICS LOCKS)

Principle. Each gate outputs: metric(s), threshold, pass/fail, and an action. A failed gate blocks the confirmatory claim.

XI.1 Gate table (locked thresholds)

Gate	Test	Threshold (locked)	Status
G0	Eligibility	$n \geq 500$, $\text{span}(X) \geq 10^{-17}$, $S \geq 2$ segments	PASS/FAIL
G1	Witness ($\Delta h \simeq 0$)	$ \hat{\epsilon}_{\Phi}^{(0)} /\text{SE}_{\text{HAC}} < 2$	PASS/FAIL
G2	LOO segments	$ \hat{\epsilon}_{\Phi}^{(-s)} - \hat{\epsilon}_{\Phi} \leq 1 \text{ SE}$ and sign stable	PASS/FAIL
G3	Exposure uncertainty	Deming/WTLS (if triggered) shifts $\hat{\epsilon}_{\Phi}$ by $\leq 0.5 \text{ SE}$	PASS/FAIL/NA
G4	Covariate veto	Adding Z shifts $\hat{\epsilon}_{\Phi}$ by $\leq 0.5 \text{ SE}$ (when Z available)	PASS/FAIL/NA
G5	Residual diagnostics	Ljung–Box $p > 0.01$ on $\geq 80\%$ of segments	PASS/FAIL
G6	Link/ops + step audit	Good-link fraction ≥ 0.90 <i>and</i> any unflagged step audit (if triggered) shifts $\hat{\epsilon}_{\Phi}$ by $\leq 0.5 \text{ SE}$ (when flags/metrics exist)	PASS/FAIL/NA
G7	Influence (points)	$\#\{i : \text{CookD}_i > 4/n\}/n \leq 0.01$ and $\max \text{DFBETA}_{\epsilon_{\Phi}} \leq 1$	PASS/FAIL

XI.2 Gate notes

- S is the number of valid segments after inclusion/exclusion.
- G1 uses the witness subset (Section V): refit the **same** primary model on $\{|X| < X_0\}$ and require the corresponding t-statistic for $\hat{\epsilon}_\Phi^{(0)}$ to satisfy $|t^{(0)}| < 2$.
- “NA” is allowed only when the gate is mathematically inapplicable (e.g. no Z provided).
- If G5 fails, GLS AR(1) sensitivity becomes mandatory to report, but does not rescue the claim (Case C still applies).
- G6 is computed only when link/ops flags or a link-quality metric is available; otherwise status = NA.
- G7 uses point-level influence diagnostics under the primary fit (Cook’s distance and DFBETA for ϵ_Φ); the full distributions are archived in the run report.
- If any gate FAILs, the confirmatory claim is blocked (Case C) and time-series robustness outputs (GLS AR(1), block bootstrap) are reported for diagnostics.

XII — BLINDING, FREEZE, AND EXECUTION ORDER

XII.1 Phase 0 — synthetic validation (mandatory)

Objective: verify the pipeline (i) recovers injected slopes with negligible bias, (ii) does not fabricate slopes when the injection is zero, (iii) passes placebo gates.

XII.2 Phase 1 — light blinding (development)

During development, only $Y^{(\text{blind})}$ and placebo/permutation outputs are visible. It is forbidden to tune the pipeline after observing any unblinded primary estimate.

XII.3 Phase 2 — freeze (exact moment)

Freeze locks everything that can change the primary outcome:

- SAP freeze: public deposition (Zenodo/OSF). Record the PDF SHA256 in `FREEZE.txt`. Also record it in `MANIFEST.json`.
- Code/config freeze: archive the exact analysis release (scripts + `config.yaml`). Record tag/commit (or Zenodo record) in `FREEZE.txt`.
- Dependencies: frozen environment file (`requirements.txt` or `renv.lock`).
- Data freeze: all sources and SHA256 listed in `MANIFEST.json`.

XII.4 Phase 3 — primary execution and unblinding

Order is non-negotiable:

1. Run `python run_all.py -blind` to generate diagnostics and gates on blind/placebo.
2. Verify integrity checks and produce the gate table.
3. Unblind (single switch) and rerun `python run_all.py -unblind` without code changes.
4. Produce the primary output: $\hat{\epsilon}_\Phi$, SE, CI, gate table, and the full report.

Every primary run produces a `RUN_REPORT.json` with versions, hashes, seeds, and results.

XIII — REPRODUCIBILITY (REPLICATION PACKAGE: ONE COMMAND)

Goal. A reviewer can rebuild all tables/figures and the primary estimate with a single command on a standard machine.

XIII.1 Required artefacts

- This SAP PDF.
- `MANIFEST.json` (sources + checksums).
- Frozen environment file(s).
- `config.yaml` (all operational parameters).
- `run_all.py` script entrypoint + `src/` modules.
- `FREEZE.txt` (timestamp + signatures + hashes).

XIV — MANDATORY OUTPUTS (TABLES/FIGURES)

XIV.1 Tables

1. T1: Source inventory (DOI/URL, version/date, checksum, available variables).
2. T2: Segment inventory (segment_id, duration, n , gaps, configuration).
3. T3: Primary result ($\hat{e}_\Phi$, SE(HAC), 95% CI, t , gates status).
4. T4: Gates (metrics + thresholds + PASS/FAIL/NA).
5. T5: Sensitivities ($\hat{e}_\Phi$, SE, CI per variant).
6. T6: Deviations (if any): type, justification, impact.

XIV.2 Figures

1. F1: Distribution of X (histogram + 0.05/0.95 quantiles).
2. F2: Y vs X scatter with primary fit and CI band.
3. F3: Residual diagnostics (residuals vs time + ACF or equivalent).
4. F4: Placebo/permutation distribution of $\hat{e}_\Phi$ with true $\hat{e}_\Phi$ marker.
5. F5: Influence plot (LOO segments: $\hat{e}_\Phi^{(-s)}$ waterfall).

XV — DEVIATIONS, ERRORS, AND AUDIT TRAIL

Rule. A deviation is not a sin. An unlogged deviation is a fault.

XV.1 Taxonomy

- Minor: does not change the primary endpoint or any gate (e.g. plot bug).
- Major: can change $\hat{e}_\Phi$, SE, CI, t , or any gate status.

XV.2 Bug handling

If a bug is discovered after freeze:

1. Document the bug with a minimal reproduction.
2. Apply a deterministic fix on a bugfix branch.
3. Produce a clear diff and quantify impact on outputs.
4. Publish an addendum. The SAP v1.0 remains immutable unless the rule itself must change (then new version).

XV.3 Forbidden deviations (zero tolerance)

- Changing a gate threshold after seeing unblinded results.
- Adding/removing sensitivities to “save” significance.
- Replacing a data source with a different one post-hoc (unless checksum identical).

XVI — GOVERNANCE AND WORKLOAD (PI/HOST-FRIENDLY)

XVI.1 Roles

- **Author/Executor (Benjamin Bréchet):** implementation, execution, writing, deposition, replication package, reviewer responses.
- **PI (to be assigned):** sanity-check of endpoints/gates/thresholds, short review of dominant systematics, validation of freeze (FREEZE.txt), quarterly check-ins (light workload).
- **Host unit:** institutional framing and administrative support as needed; no technical ownership required.

XVI.2 Cadence

- Quarterly 30-min check-in: gate status, systematic risks, next steps.
- M3: pipeline + placebos + synthetic validation complete.
- M6: freeze + primary execution.
- M9–M12: manuscript + public replication package.

XVII — DECLARATIONS

Availability

This SAP (v1.0) is published as an immutable preregistration artifact. The PDF checksum and build metadata are provided in the replication package (FREEZE.txt + MANIFEST.json inside this ZIP release).

The main protocol is referenced by DOI: 10.5281/zenodo.17571491. If "sources" is empty at publication time, it means: NOT EXECUTED; primary execution requires completing MANIFEST.json with pinned sources (DOI/URL + version/date + SHA256) before unblinding.

Data

This SAP enforces exclusive use of public/archived datasets with checksums (Section IV). No personal or sensitive data.

Conflicts of interest

None.

XVIII — REFERENCES (MINIMAL)

1. B. Bréchet, *CHRONON-1: Experimental Pre-Registration (program protocol)* (v1.0). Zenodo. DOI: 10.5281/zenodo.17571491.
2. B. Bréchet, *CHRONON-1 — SAP/Prereg replication package* (this ZIP release: CHRONON-1_SAP-Prereg_Replication-Package_v1.0_zenodo-18072280.zip). Zenodo DOI assigned upon deposition.
3. W. K. Newey and K. D. West, "A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix", *Econometrica* 55(3), 703–708 (1987).
4. D. W. K. Andrews, "Heteroskedasticity and autocorrelation consistent covariance matrix estimation", *Econometrica* 59(3), 817–858 (1991).
5. G. M. Ljung and G. E. P. Box, "On a measure of lack of fit in time series models", *Biometrika* 65(2), 297–303 (1978).

XIX — APPENDICES (TEMPLATES)

Annex E — MANIFEST.json (traceability contract)

Template

```
{
  "sap_version": "v1.0",
  "sap_sha256": "see FREEZE.txt",
  "protocol_doi": "10.5281/zenodo.17571491",
  "exec_package_doi": null,
  "created_utc": "<UTC_TIMESTAMP>",
  "sources": [
    {
      "test": "A",
      "name": "dataset_clock_1",
      "doi_or_url": "<DOI_OR_URL>",
      "version_date": "<YYYY-MM-DD>",
      "filename": "<FILENAME>",
      "sha256": "<SHA256_HEX>"
    }
  ],
  "notes": "public-only metadata; no secrets"
}
```

Note. Keep exec_package_doi as null until Zenodo assigns a DOI to the ZIP release.

Annex F — config.yaml (locked operational parameters)

Template

```
seed_global: 1729
seed_blind: 31415
```

```
# Segmentation (Section VI)
tau_gap_hours: 24
Tmin_hours: 48

# Gate thresholds (Section XI)
n_min: 500
span_X_min: 1.0e-17
lb_p_min: 0.01
lb_pass_frac: 0.80
link_good_frac_min: 0.90
cookd_frac_max: 0.01
dfbeta_abs_max: 1.0

# Change-point audit (Gate G6)
step_mad_kappa: 8
step_shift_se_max: 0.5

# Outliers
mad_kappa: 5

# Outputs
out_dir: "reports/"
```

Annex D — Freeze-ready checklist

Checklist

1. SAP PDF built; SHA256 recorded in `FREEZE.txt` and `MANIFEST.json`.
2. Replication package archived (this ZIP release); identifier recorded in `FREEZE.txt`.
3. `MANIFEST.json` complete and checksums validated.
4. `config.yaml` archived; no “TBD” parameters.
5. Synthetic validation passed (Phase 0 logs archived).
6. `run_all` rebuilds T1–T6 and F1–F5 in one command.
7. Signatures collected (you + PI if applicable) on `FREEZE.txt`.